Eduardo Castro

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Education

Union College, Schenectady, NY — Expected June 2027
Bachelor of Science: in Mechanical Engineering — Cumulative GPA: 3.97
Bachelor of Arts: in Astronomy
Honors: Hadala Prize (2026), Davenport Fellowship, Dean's List 2023-25, Tau-Beta-Pi (Treasurer), Sigma-Pi-Sigma
Relevant Course Work: Advanced Manufacturing, Aerospace Structures, Orbital Mechanics, Heat Transfer

Skills

Software: CAD/FEA(Siemens NX, SolidWorks, Creo, Fusion360), CFD(STAR-CCM+), Microsoft Office, Google Suite
Programming Languages: MATLAB, Python, Arduino, Java
Technical: Milling, Lathe, 3D Printing, Arduino, Raspberry Pi, GD&T, Laser Use and Safety
Language: Portuguese (Fluent)

Experience

Mechanical Engineering Intern, SunThru (Silica Aerogel Startup) – Schenectady, NY — Apr 2026 – Present

- Design experimental systems and operate silica-aerogel production presses; report directly to the CTO
- Built HELIOS, a company-wide lab platform: digital notebook, analysis pipeline (120+ metrics), and AI assistant
- Identified statistical drivers of product quality; led electrostatic-defect investigation; mitigations adopted
- Authored 15+ standard operating procedures; designed 3D-printed test fixtures and new sensor instrumentation

Fluid Dynamics Research Assistant, Union College – Schenectady, NY — June 2025 – Present

- Design and conduct wind-tunnel/water-channel PIV experiments on heating effects in turbulent rough-surface flow
- Operate high-powered laser and high-speed imaging systems; analyze large flow datasets with custom MATLAB code
- Presented at the 2026 Steinmetz Symposium (ASME IMECE 2026 poster in prep); mentored a high-school researcher

President, Union College Rocket Team – Schenectady, NY — Sept 2024 – Present

- Re-founded competition team; grew membership from 5 to 18 and budget from $500 to $3,850
- Led 12-member cross-disciplinary team to 1st place in the Deployable Payload Event at the 2026 Battle of the Rockets national competition, presenting Testing and Logistics at design reviews (CDR: 89%)
- Organize and enforce safety procedures in design, handling, and use of solid rocket fuel and black powder charges

Lead Thermal Control Engineer, L'SPACE Mission Concept Academy, NASA Lucy Mission – Remote — Jan 2025 – Apr 2025

- Designed the Thermal Control Subsystem for a theoretical Venus atmospheric aerobot: requirements definition, trade studies, component selection, and verification plans for a 475°C, high-pressure environment
- Produced Interface Control documentation, Siemens NX CAD assemblies, and N^2 diagrams for the 12-person team

Scholars Astrophysics Research Team Member, Union College – Schenectady, NY — Sept 2023 – June 2025

- Operated Union College Observatory to obtain, analyze, and publish photometric data on asteroid rotation
- Co-authored five Minor Planet Bulletin papers (first author on one) and three TESS exoplanet submissions (TFOP)
- Presented at the Astronomical Society of New York, NY6, and Steinmetz Symposium conferences (2024-25)

Projects

Modular Horizontal Axis Wind Turbine

- Designed and built a 5 ft-tall, 3 ft-diameter wind turbine with custom 3D-printed components and DC generator
- Optimized performance via SolidWorks FEA and experimentally for varying chord lengths and angles of attack